

LED Dial-Tuning Manufacturers — PGM/REE Metallurgical Processing

Hex rotary dial device (0–F wavelength × 0–F power, 100 mW steps to 1.5 W) mapped to closest commercial LED systems for Platinum Group Metal and Rare Earth Element photonic processing.

Hex Dial Wavelength Decode

Dial	Wavelength	PGM/REE Target	Match Quality
0	PROG	Custom sequence	—
1	280 nm	Pt secondary, Gd	Good
2	367 nm	Tb ³⁺ (365nm), Sm, Ag photolysis	±2 nm Excellent
3	405 nm	Ag SPR, Eu ³⁺ , Pd porphyrin	Exact
4	448 nm	Ru(bpy) ₃ ²⁺ , Ho ³⁺	±2 nm Excellent
5	470 nm	Nd adjacent (464 nm)	±6 nm Good
6	505 nm	Tb emission verify	Good
7	530 nm	Au SPR (peak 520 nm)	±10 nm Good
8	568 nm	Tb emission (546 nm)	±22 nm Acceptable
9	591 nm	Eu emission (594 nm)	±3 nm Excellent
A	617 nm	Eu ³⁺ emission	Exact
B	627 nm	Eu/Pr range	Good
C	655 nm	Er ³⁺ excitation (660 nm)	±5 nm Good
D	740 nm	NIR band	Good
E	850 nm	NIR broad	Exact
F	940 nm	NIR	±40 nm Good

★★★★★ BEST MATCH — CombiLED mirrors dial device — Prizmatix — Israel

Product: CombiLED / UHP-F-LED Series

CombiLED combines up to 8 LED wavelength modules into a single fiber output with I²C microcontroller control — identical architecture to the dial device. Wavelengths 280–1100 nm match nearly every dial position. 12-bit power resolution per channel. Recommended 8-channel PGM/REE config: 280, 365, 405, 450, 530, 617, 660, 850 nm.

Website: <https://www.prizmatix.com/combled/combled.htm>

Contact: sales@prizmatix.com

Module	λ (nm)	Max Power	Dial
UHP-F-280	280	300 mW	Pos 1
UHP-F-365	365	1000 mW	Pos 2 ✓
UHP-F-405	405	1500 mW	Pos 3 ✓
UHP-F-450	450	2000 mW	Pos 4 ✓
UHP-F-530	530	800 mW	Pos 7 ✓

UHP-F-617	617	600 mW	Pos A ✓
UHP-F-660	660	1000 mW	Pos C ✓
UHP-F-850	850	500 mW	Pos E ✓

★★★★★ LIKELY ACTUAL DEVICE — Thorlabs M-series LEDs — Thorlabs — USA

Product: M-series Mounted LEDs + DC4100 / LEDD1B Drivers

Thorlabs M-series mounted LED wavelengths match dial positions 1–F almost exactly. The reference device is likely a custom controller board driving Thorlabs LEDs. DC4100 4-channel driver (I²C/USB) or LEDD1B single-channel drivers (1.2 A each). Full catalog: 265–1700 nm.

Website: https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=2692

Contact: techsupport@thorlabs.com

Part #	λ (nm)	Power	Dial pos
M280LP1	280	80 mW	Pos 1
M365LP1	365	1200 mW	Pos 2
M405L4	405	1300 mW	Pos 3
M470L5	470	1300 mW	Pos 5
M530L4	530	700 mW	Pos 7
M617L3	617	700 mW	Pos A
M660L4	660	700 mW	Pos C
M850L3	850	600 mW	Pos E
M940L3	940	500 mW	Pos F

★★★★★ CLOSE SECOND — Mightex Systems — Canada

Product: WFC-BLS Multi-Wavelength LED

Up to 8 LEDs combined in single fiber output, I²C/USB, 280–1100 nm, 12-bit DAC. Same concept as Prizmatix with slightly fewer catalog options.

Website: https://www.mightexsystems.com/family_info.php?cPath=24_51

Contact: info@mightexsystems.com

Series	Channels	Range	Control
WFC-BLS	2–8	280–1100 nm	I ² C/USB
SLC-AA	1	340–1100 nm	RS-232

★★★ HIGH-END 32-CHANNEL — Gamma Scientific — USA

Product: SpectralLED RS-7-VIS / RS-7-UV

32-channel LED array, 16-bit DAC per channel, SCPI/RS-232. RS-7-VIS covers 380–1000 nm; RS-7-UV covers 200–400 nm. Lab reference standard — more channels than the dial device, same philosophy.

Website: <https://www.gamma-sci.com/products/spectraledlightsources/>

Contact: sales@gamma-sci.com

Model	Channels	Range	Control	DAC
RS-7-VIS	32	380–1000 nm	SCPI	16-bit

RS-7-UV	32	200–400 nm	SCPI	16-bit
RS-7-IR	32	700–1700 nm	SCPI	16-bit

★★★ DEEP UV GAP FILL — Pt/Os/Ir primary lines — Crystal IS / SETi (Asahi Kasei)

Product: Optan Series / UVTOP265 AlGaIn UV-C LEDs

AlGaIn UV-C LEDs for deep UV — the wavelength gap the hex-dial device cannot cover. UVTOP265 at 265 nm hits Pt primary line exactly. Must supplement dial device with these for complete Pt/Os/Ir coverage.

Website: <https://www.crystal-is.com/products/optan-series/>

Contact: sales@crystal-is.com

Part	λ (nm)	Power	PGM target
UVTOP265	265	1–5 mW	Pt primary (265.9 nm)
UVTOP255	255	1–3 mW	Ir primary
Optan-270	270	10 mW	Pt/Os region
Optan-250	250	5 mW	Os/Ir region

★★★ NIR GAP FILL — Nd/Yb/Er pump — Roithner LaserTechnik — Austria

Product: ELJ-808 / ELJ-980 NIR Laser Diodes

NIR laser diodes covering 808 nm (Nd³⁺ pump) and 980 nm (Yb³⁺/Er³⁺ pump) gaps in the hex dial device. ELJ-808-500M is a 500 mW pigtailed diode.

Website: https://www.roithner-laser.com/laser_diodes_808.html

Contact: office@roithner-laser.com

Part	λ (nm)	Power	REE target
ELJ-808-500M	808	500 mW	Nd ³⁺ pump
ELJ-980-100	980	100 mW	Yb ³⁺ /Er ³⁺
ELJ-975-200	975	200 mW	Yb ³⁺ exact
ELJ-730-100	730	100 mW	Nd ³⁺ alt

All hyperlinks active — tap/click to open. Cross-referenced: STARE Research Division PDF June 25, 2026.